

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (EC)

Sem - III (Backlog)

University Examination, 2010-11

EC – 202 Network Theory

Date: 18.05.11, Wednesday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION -I

- Q-1(a) Differentiate active and passive networks. 3
(b) Determine the current through the ammeter of resistance 2Ω connected in the wheatstone bridge of **Figure 1**. 5
(c) Define the following 4
(i) Network (ii) Bode Plot (iii) Nyquist Criterion (iv) Magnitude Plot
Q-2(a) Explain Initial conditions for different elements 6
(b) An exponential voltage $v(t)=4e^{-3t}$ is applied at time $t=0$ to a series R-L circuit consisting of a resistor $R=0.5\Omega$ and inductor $L=0.25H$ shown in **Figure 2**. Obtain the particular solution for current $i(t)$ through the circuit. Assume zero current through the inductor at time $t=0$. 6

OR

- Q-2(a) Explain solution of homogeneous and non-homogeneous differential equations. 6
(b) In the given network of **Figure 3** switch k is opened at $t=0$. Solve for v , dv/dt and d^2v/dt^2 at $t=0+$ if $I=10A$, $R=10\Omega$ and $L=1H$. 6
Q-3(a) Find the Laplace transform of an isosceles triangular pulse shown in **Figure 4**. 6
(b) Solve the following differential equation using Laplace Transformation method, 5

$$\frac{d^2i}{dt^2} + 4\frac{di}{dt} + 5i = 5u(t)$$

Given that $i(0+) = 1$ and $di/dt(0+) = 2$.

OR

- Q-3(a) The waveform shown in **Figure 5** is a terminated truncated ramp. Determine the Laplace transform $V(s)$ 6
(b) Find the inverse Laplace transform of the following function. 5

$$F(s) = \frac{s-1}{s^2 + 3s + 2}$$

SECTION -II

- Q-4(a) Explain relationship between ABCD parameters and z-parameters. 6
 (b) Define KCL, KVL . Explain concept of Súpermesh 5

- Q-5(a) Explain h-parameters. 6
 (b) Obtain the z-parameters for the network of **Figure 6**. 6

OR

- Q-5(a) Explain relationship between h-parameters and g-parameters 6
 (b) Find the y-parameters for the resistive network of **Figure7**. 6

- Q-6(a) Explain source transformations in detail 6
 (b) Find out the solution using $e^{\pm j\omega t}$ for the series R-L circuit shown in **Figure 8**. 6

OR

- Q-6(a) Explain Phasors and Phasor diagrams 6
 (b) Find out the solution using $e^{\pm j\omega t}$ for the parallel R-L circuit shown in **Figure 9**. 6

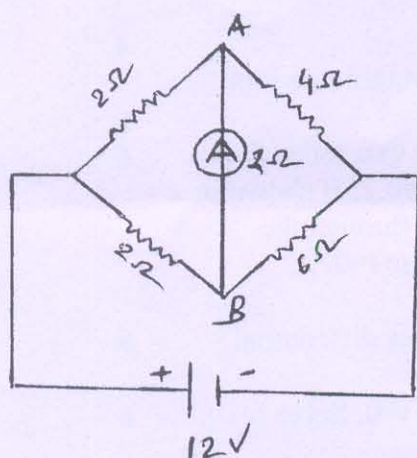


Figure - 1

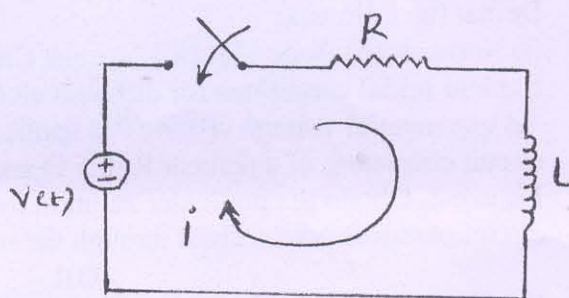


Figure - 2

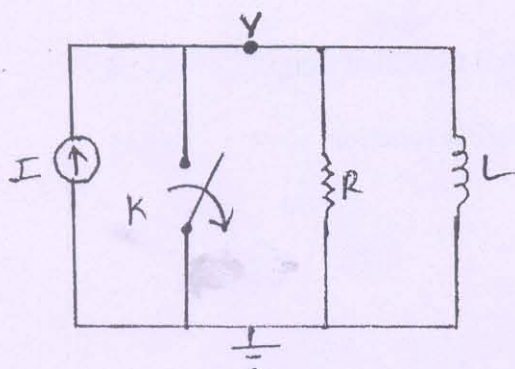


Figure - 3

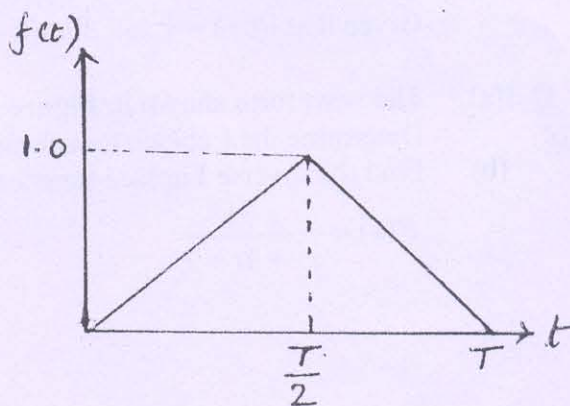


Figure - 4

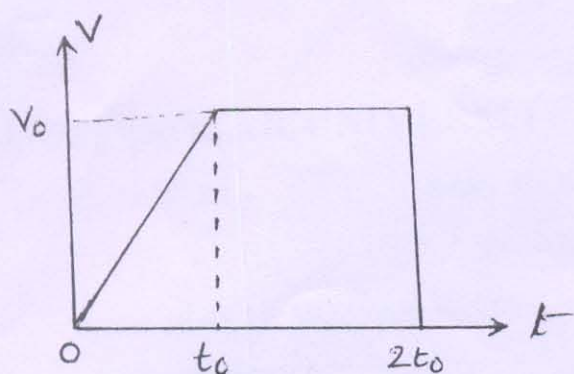


Figure-5

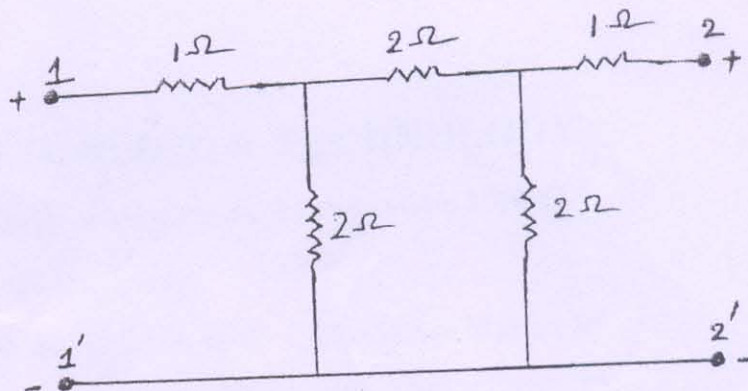


Figure-6

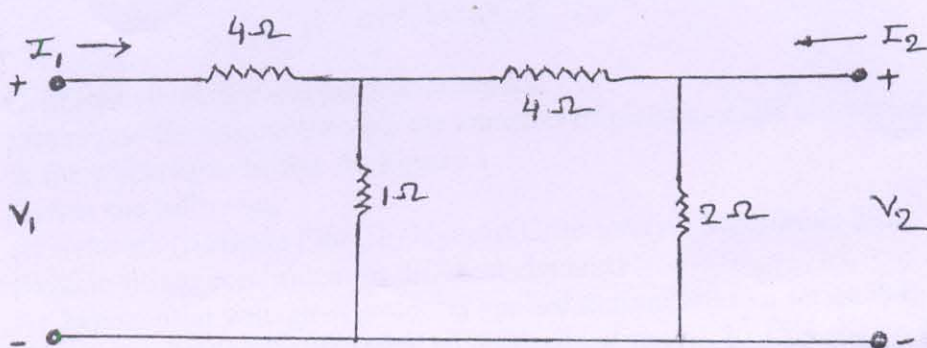


Figure-7

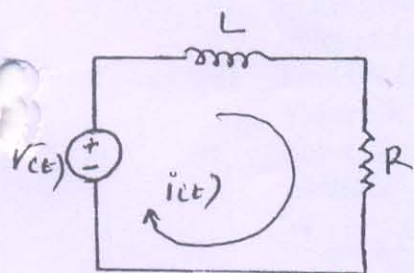


Figure-8

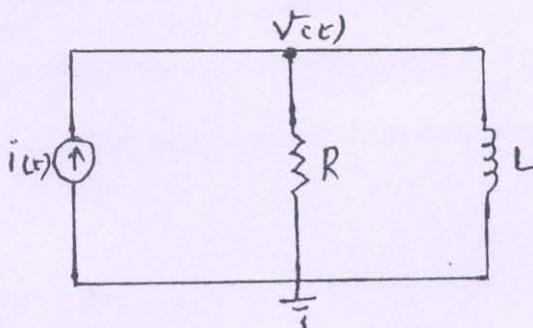


Figure-9